

Federica Padovano

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Email: federica.padovano@sorbonne-universite.fr

Memberships: SMB, ESMTB

Office: 4 Pl. Jussieu, Paris, 75005, France

Languages: Italian, English, French

Education

2023-present	LJLL, Sorbonne Université (SU) PhD in Applied Mathematics. <u>Title:</u> “Mathematical modelling of cancer therapies “ <u>Advisors:</u> Prof. L. Almeida (LPSM, SU), Prof. M. Cerezo (Université Côte d’Azur), Dr. C. Villa (MAP5, UPC)
2021-2023	École Polytechnique Fédérale de Lausanne (EPFL) MSc in Applied Mathematics. <u>Master project:</u> “An evolutionary dynamics model of the development of drug resistance in metastatic tumors“ (LJLL, Sorbonne Université, Paris) <u>Advisors:</u> Prof. L. Almeida, Prof. M. Picasso (EPFL), Dr. C. Villa
2018-2021	Politecnico di Torino (PoliTo) BSc in Mathematics For Engineering. <u>Bachelor project:</u> “Study of an aggregation and propagation model of amyloid in neurons to describe the progression of Alzheimer’s disease” <u>Advisor:</u> Prof. A. Tosin (PoliTo)

Industry experience

09/2022-02/2023	Novadiscovery, Lyon Biomodeler internship, “Physiologically-based pharmacokinetic model for small molecule drugs”
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Publications & Preprints

- F. Padovano, C. Villa, *The development of drug resistance in metastatic tumours under chemotherapy: an evolutionary perspective*, Journal of Theoretical Biology, 595(1):111957, 2024, <https://doi.org/10.1016/j.jtbi.2024.111957>.
- R. Singh, J. Glowacki, M. Beaud, F. Padovano, R.S. Manning, J.H. Maddocks, *cgNA+min: computation of sequence-dependent dsDNA energy-minimizing mini-circles*, Nucleic Acids Research, 54(9), 2026, <https://doi.org/10.1093/nar/gkag398>
- F. Padovano, , A. M. Lombardi (UniTo), S. Novello (UniTo), T. Lorenzi (PoliTo), E. Vigna (UniTo), E. Leschiera (ESILV), C. Villa, *Modelling the efficacy of CIK cell immunotherapy targeting MET-expressing mesothelioma cells* (preprint available on ResearchGate)

Ongoing projects

F. Padovano, L. Almeida, A. C. Bastos (Université Côte d'Azur), M. Cerezo, C. Villa, *Mathematical modelling of the role of nutrients in the tumour microenvironment on the efficacy of melanoma treatment*

Meetings with contribution

06/2026, Paris (LJLL)	Workshop on PDEs and Mathematical Biology (Workshop - Invited Speaker)
06/2026, Nanterre	Collective Behavior in Complex Systems (Conference - Keynote Invited Speaker)
05/2026, Oxford	Mathematical Oncology Group meeting (Visiting period - Invited speaker)
02/2026, PoliTo	From Micro To Macro in Mathematical Biology: Recent advances in the modeling and analysis of biological systems (Workshop - Contributed talk)
11/2025, Grenoble	J-Biot 2025: Les 1ères Journées de la Biologie Théorique (Workshop - Contributed talk)
09/2025, SIMAI	Mathematical and Computational Models for Tumor Evolution and Therapy (Conference minisymposium - Invited speaker)
06/2025, SMAI	Mathematical advances in modelling cancer treatment (Conference minisymposium - Invited speaker)
04/2025, Bonn	Young women in mathematical biology (Workshop - Invited speaker)

Future appointments

07/2026, ECMTB	Deterministic models to describe and analyse tumour-immune interactions and immunotherapy (Conference minisymposium organiser and speaker)
10/2026, Paris (LJLL)	Data-driven approaches in modelling cell and tissue dynamics (Workshop organiser)

Grants

04/2025	European Women in Mathematics (EWM) travel grant (400€)
10/2025	BOUM SMAI funding for workshop organisation in 09/2026 (500€)
04/2026	European Society of Mathematical and Theoretical Biology (ESMTB) travel grant (350€)
06/2026	BioConvergence pour Santé (BioConvS) funding for workshop organisation (1000€)

Teaching experience

2023-2026	Teaching assistant, Sorbonne Université • 54h (1 semester) of the BSc course <i>Mathematics for sciences 1</i> (Basic notions: calculus, differential equations, multivariable functions, and linear transformations of the plane) • 138h (3 semesters) of the BSc course <i>Mathematics for sciences 2</i> (Linear algebra and Probability)
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